Reviewer 1:

Overall merit

1.
Reject

Reviewer expertise

5.
I am an expert on this topic (e.g., I work/have published on this topic)

Paper summary

This paper explores the explainability of issue reports from GitHub-based Projects. By comparing the faithfulness and interpretability of LIME and SHAP methodologies, they applied seBERT and IssueBERT for multi-class issue classification of three categories: Bug, Enhancement, and Question. They have used two faithfulness-based metrics- Sufficiency and Comprehensiveness. Their results show that IssueBERT outperformed seBERT for the issue classification task. For explainability, they found that the top 10 tokens are better than the top 5 or top 20 tokens for both LIME and SHAP. Faithfulness metrics stabilize at a small number of key tokens, highlighting an efficient balance between interpretability and explanation of completeness. This work would be a foundation for future explainable AI research in the Software Engineering domain.

Strengths and Weaknesses

Here are the strong points of the paper: +important and interesting problem for SE

- large-scale evaluation

The following points should be improved (Weakness):

- some wrong/improper methodology
- missing qualitative analysis for SHAP and LIME
- poor writing quality, and the paper is not well shaped
- missing discussion and implication of this work

Comments for authors

I would like to thank the authors for choosing a very important and interesting topic for the Software Engineering (SE) domain- explainability of AI models for SE tasks. I have shared my detailed evaluation below:

Soundness

The major issue of this paper is the lack of soundness and framing the problem. To make this paper stronger, the authors should follow proper methodology and data selection.

Abstract A summarized result is missing in the abstract. Please add it.

Introduction: In the first paragraph of the Introduction, the authors can refer to some prior works on bug reports and their classification to make a stronger motivation. Authors mentioned

“However, there are very few studies that apply such XAI methods to large-scale issue datasets involving multi-class issue classification” (line 73-75) and then started with:

“In this paper, we apply XAI techniques to the automatic classification results of 21,000 issue reports……………..”. I am missing some motivation for doing this work. You should add some concrete gaps for doing this work.

Before describing RQ1, you started to write the RQ1 findings (line 87). Same for RQ2, RQ3, and RQ4, where they did not describe the research questions first. As a reader, I would suggest that the authors describe the RQs first, with their motivation, and then write a summary of findings in the introduction.

Section 3 and 4 shoud be merged: It seems both sections 3 & 4 discusses the methodology. I would suggest merging them.

Section 3.2: The authors selected the dataset from a paper that was published in 2019.

- Rafael Kallis, Andrea Di Sorbo, Gerardo Canfora, and Sebastiano Panichella.

1. Ticket tagger: Machine learning driven issue classification. In 2019 IEEE

International Conference on Software Maintenance and Evolution (ICSME). IEEE,

406–409.

Although it is okay to use an existing benchmark dataset, a comparatively new dataset would be better. They did not properly mention why they selected this dataset. Also, the random selection of 7000 issues per class should be validated.

Model tuning The paper claimed the fine-tuning of seBERT and IssueBERT; however, they did not mention the process of fine-tuning.

Qualitative Analysis This paper uses quantitative analysis, but a qualitative analysis is required for a better understanding of the used model. I suggest that the authors do a qualitative analysis of a small part of the data.

I would suggest that authors introduce the RQs in the introduction. I am not well motivated by the following RQs:

RQ2. How do the explainability evaluation metrics change according to the number of top-k tokens ($k \in 5, 10, 20$)?

Could you please provide the motivation behind investigating different tokens?

Experimental Results

- Line 478: “For the evaluation, two datasets were used. Dataset A included 80,518 issue reports, while Dataset B included a balanced dataset (7,000 samples per class).”

I am not sure which one is dataset A. Could you please clarify?

- Analysis of RQ2 looks good but lacks of rigor. They claimed “In issue classification explainability techniques, using the top 10 tokens provides the most balanced trade-off between Comprehensiveness and Sufficiency.”

Is it true always, or only with your own dataset? Please rephrase the claim in the paper.

-Choosing the metric Comprehensiveness and Sufficiency is motivating for the evaluation.

-Figures 5 and 6 look good; however, they are not as useful to put as much space in the manuscript. Authors can move some of them to the replication package and provide more details on their findings.

- I would say the same for Figure 7, which is unnecessary to show the top 15 keywords. For RQ4, the authors showed the top 15 keywords in the figure and discussed them later. Moreover, I am not sure how this RQ helps the SE research community. Authors should motivate it properly.

Missing Discussion and Implication Section Authors should add a section with a discussion of their results and how the SE community can benefit from this study.

Novelty

Authors mentioned the novelty with the following quote: “However, our study differs from theirs in that we examine the explainability of seBERT and IssueBERT from multiple perspectives.”

I was wondering if the authors would clarify the main novelty and contribution of the paper. Although this seems novel, it lacks a contribution to the SE domain.

Relevance

The work is related to the MSR and SE domains. The following related works may help authors frame their motivation strongly.

- Cao, Sicong, Xiaobing Sun, Ratnadira Widyasari, David Lo, Xiaoxue Wu, Lili Bo, Jiale Zhang et al. "A Systematic Literature Review on Explainability for ML/DL-based Software Engineering." ACM Computing Surveys 58, no. 4 (2025): 1-34.
- Usmi Mukherjee and M. Masudur Rahman. Understanding the Impact of Domain Term Explanation on Duplicate Bug Report Detection. In Proceeding of The 29th International

Conference on Evaluation and Assessment in Software Engineering (EASE 2025), pp. 12, Istanbul, Turkey, June 2025

- Long, Guoming, Jingzhi Gong, Hui Fang, and Tao Chen. "Learning Software Bug Reports: A Systematic Literature Review." ACM Transactions on Software Engineering and Methodology (2025).

Presentation

Most of the writing of this paper is grammatically correct; however, the framing is not well structured. Please follow my above comments on the other sections.
Also, here are some issues to be fixed:

- 2.1: The first and second paragraphs will be merged.
- Figure 1 needs to be clearer and better organized
- 3.1 XAI Techniques in Issue Classification: It is better to use “Explainable AI” instead of XAI on the heading. Authors can use abbreviations in the paper.
- Some of the citations (i.e., 7, 13) are from arxiv. If possible, please refer the published version

Replicability

In the attached replication package, they have two versions, and they provided a Python code. There is no dataset attached. Apart from that, they did not share any README file or dataset.

Artifact Assessment

2.

The artifacts are present but **not** sufficiently in line with what is declared in the submission form and in the paper

Questions for authors' response

Q1. How did you fine-tune seBERT and IssueBERT? What is the train: validation: test ratio?

Q2. Did you qualitatively analyze the output of the used models? If not, how can you claim the validation of those explanations?

Comments for PC(hidden from authors)

Although this is an interesting topic, this paper lacks methodological rigor, experimentation, and presentation.